BurstiVAR for Capturing Multilevel Vector Autoregressive Processes with Trends in Measurement Burst Designs

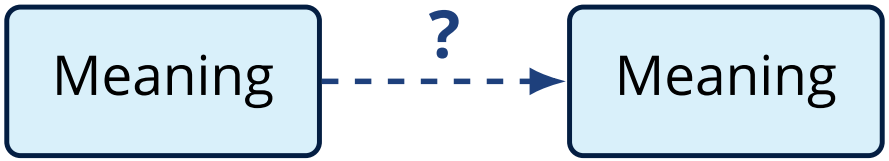
Xiaoyue Xiong[1], Ivan Jacob Agaloos Pesigan[1], Zita Oravecz[1], Stephanie T. Lanza[2], Sy-Miin Chow[1]

[1]Human Development and Family Studies; [2]Biobehavioral Health
The Pennsylvania State University

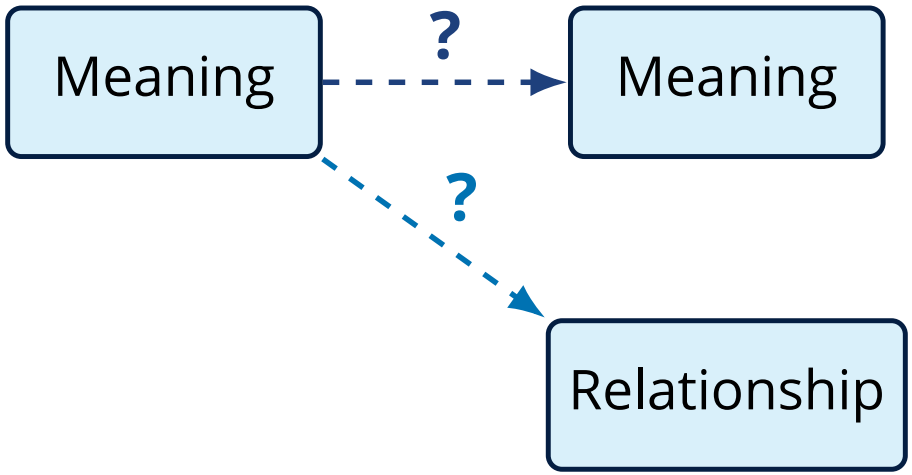
Updated 10/19/2026



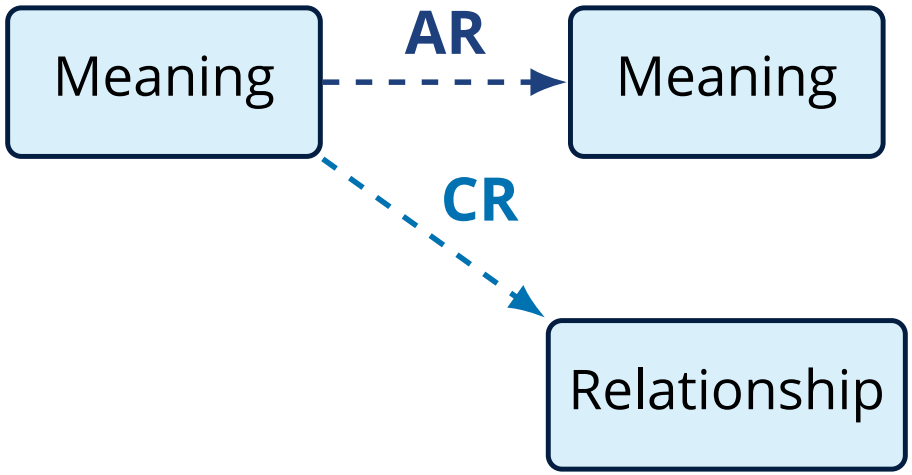
Dynamics in Short-Burst Designs



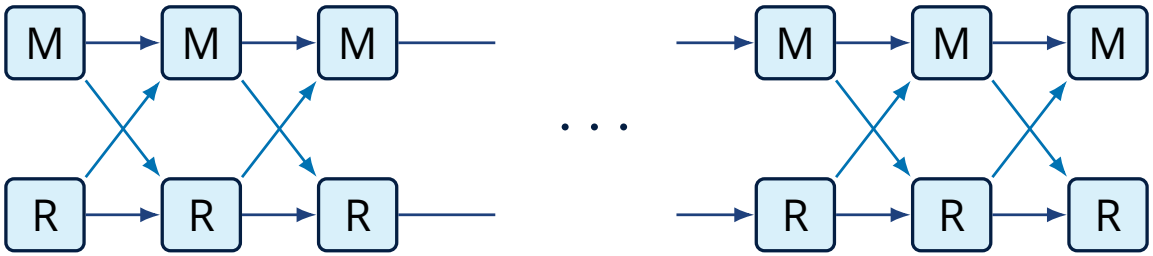
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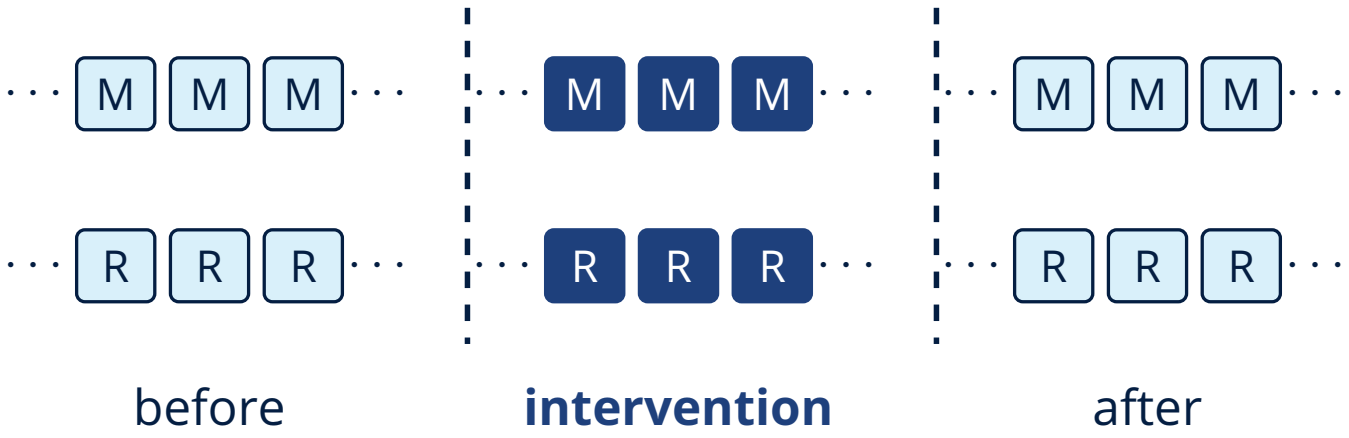


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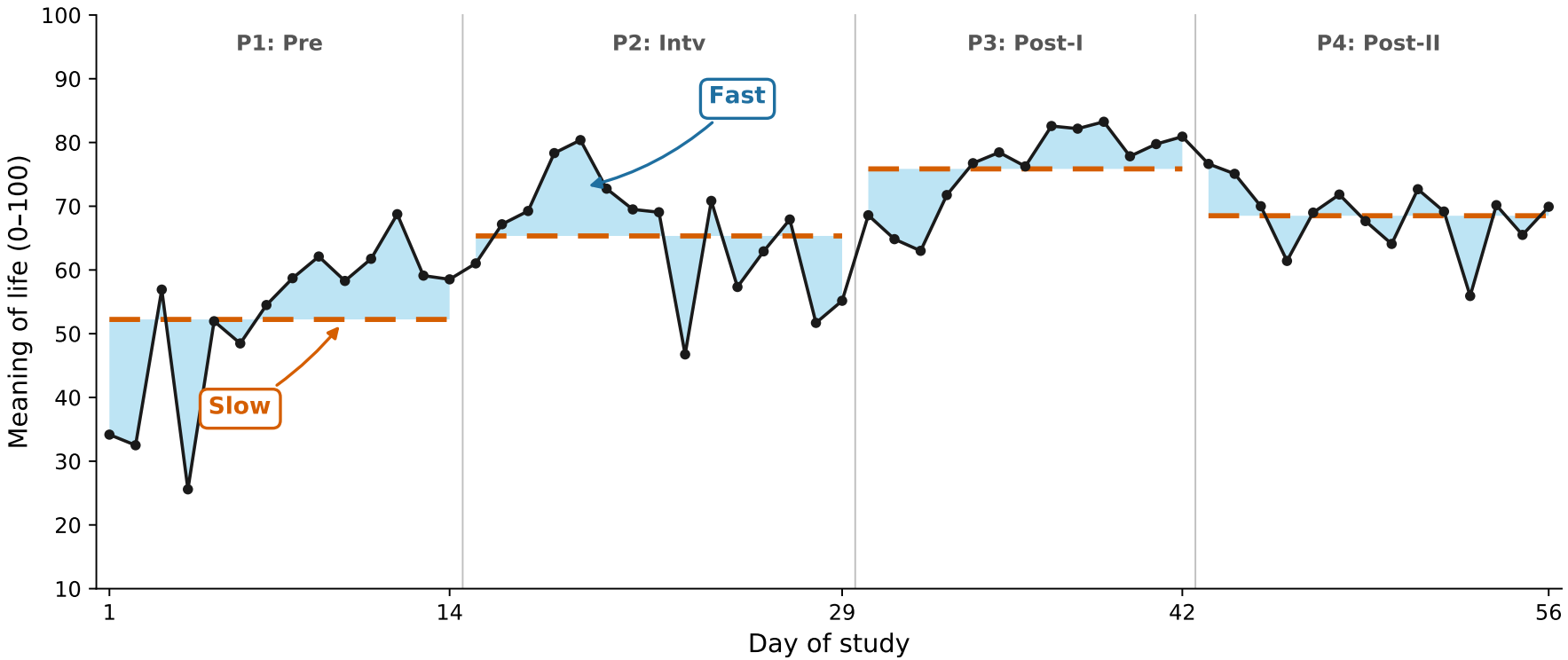
The tools want a **long** series per person…

Dynamics in Short-Burst Designs



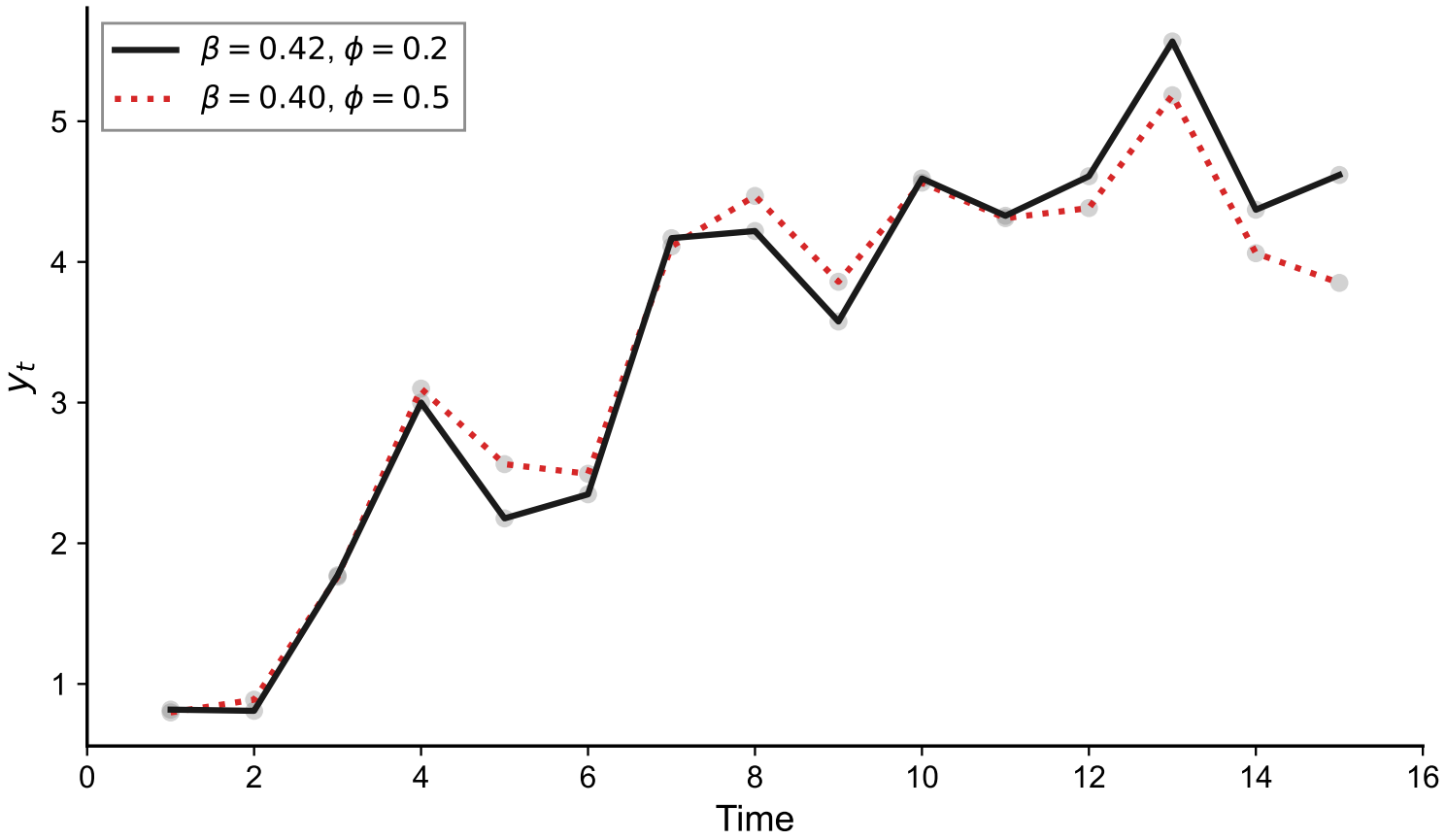
But I'd like to **graduate T T** and my participants would like their evenings back... When are there **too few points?**

Two Time Scales in One Series



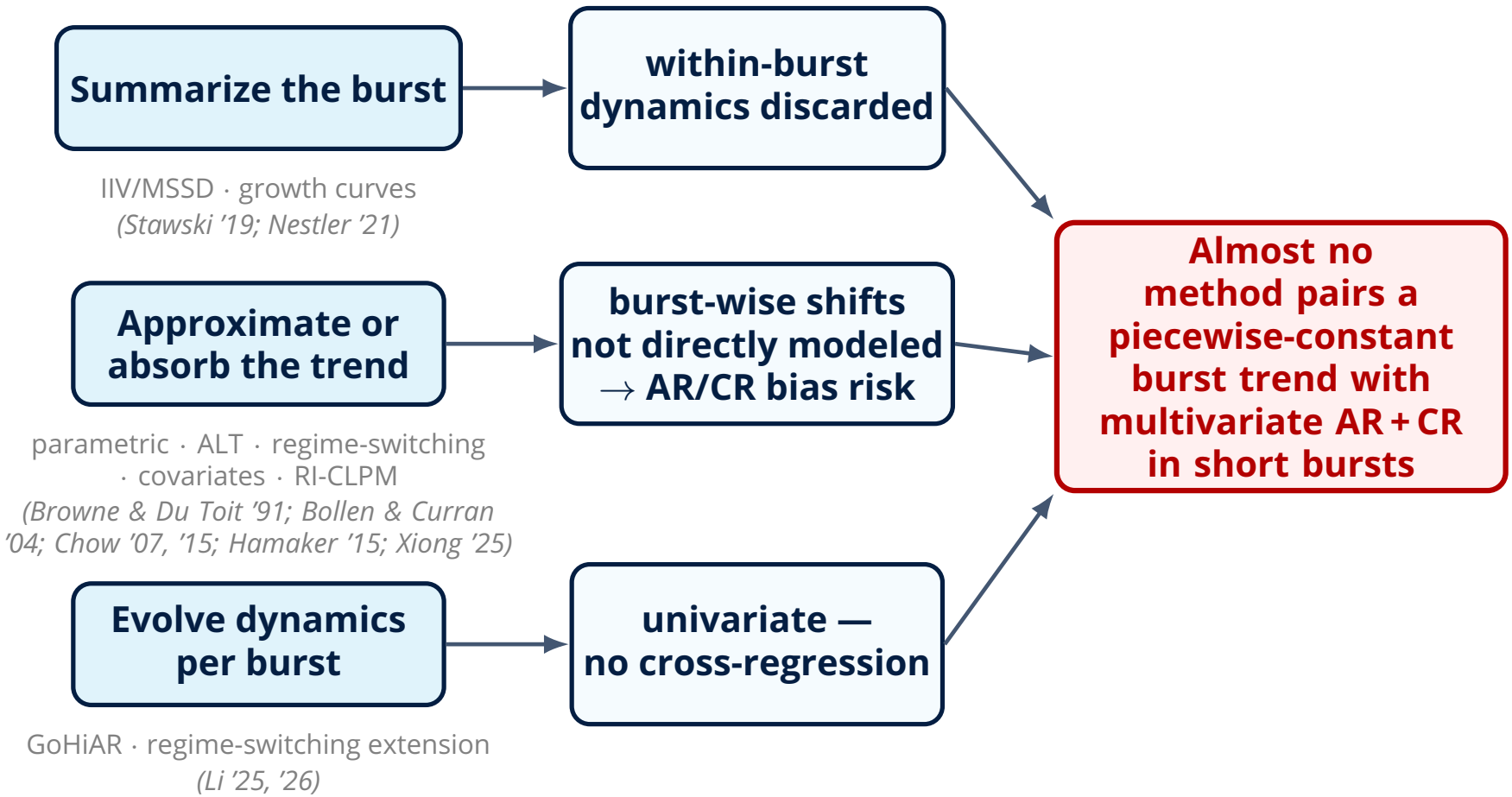
- **(Slow) Trend:** burst-to-burst drift (P1 $\rightarrow$ P4)
- **(Fast) Dynamics:** short-term carry-over *within* each phase

The Confounding of Trends and Dynamics



- Steep **trend** + fleeting **Dynamics** ≈ shallow **trend** + sticky **Dynamics**
- Hard to separate them in short series

Existing Approaches



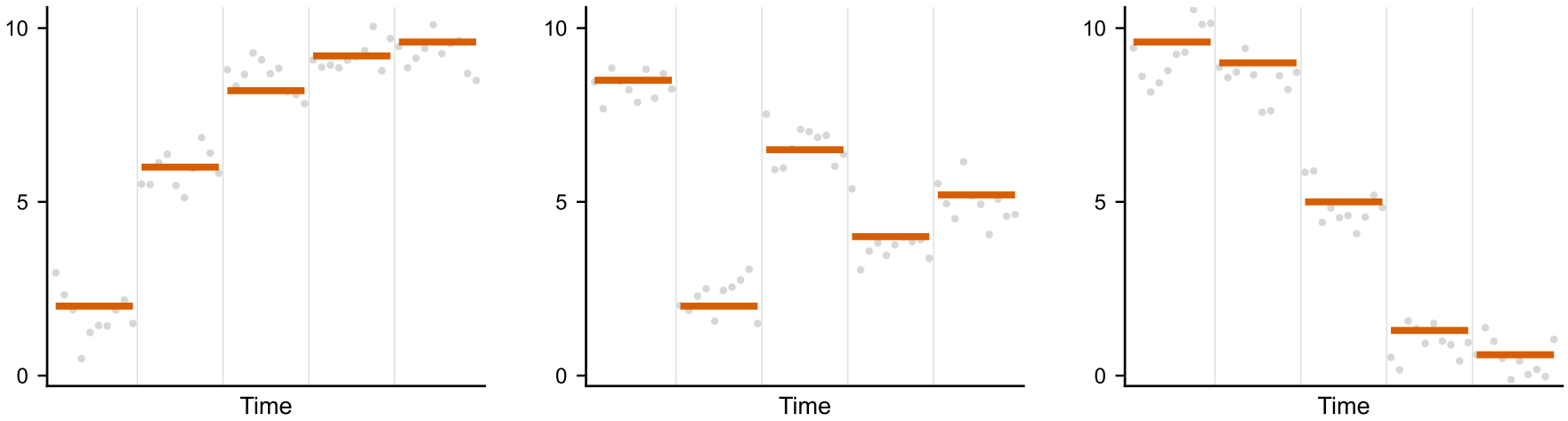
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$$\boldsymbol{y}_{ti} = \underbrace{\boldsymbol{\alpha}_{bi}}_{\text{Trend}} + \underbrace{\boldsymbol{\eta}_{ti}}_{\text{Dynamics}}$$

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α_{bi} : **Stepwise function across bursts** — *any shape*

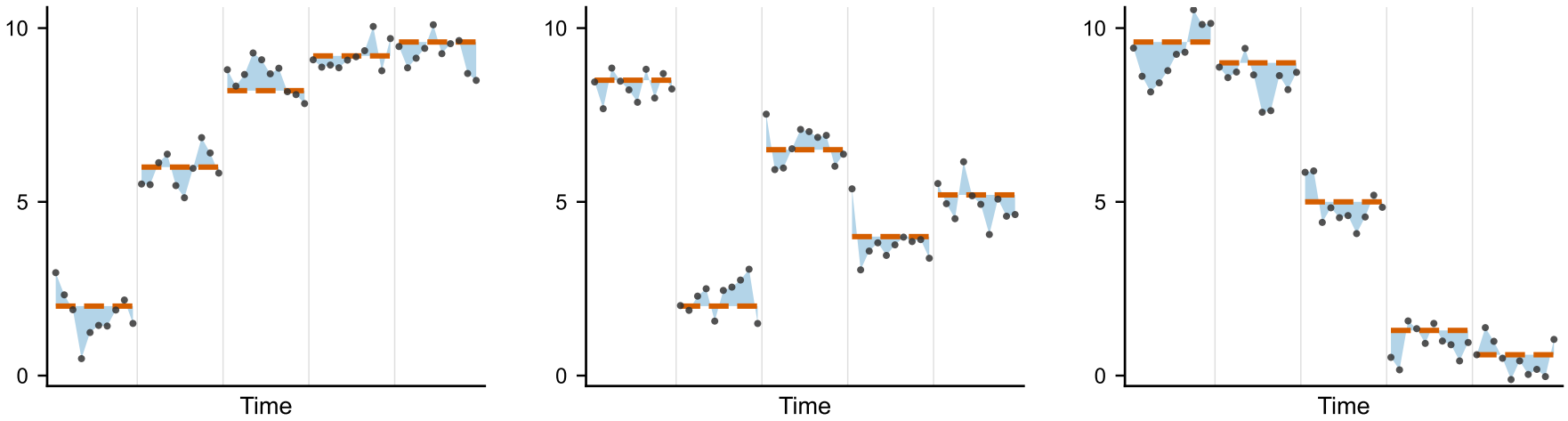


Person 1 Person 2 Person 3

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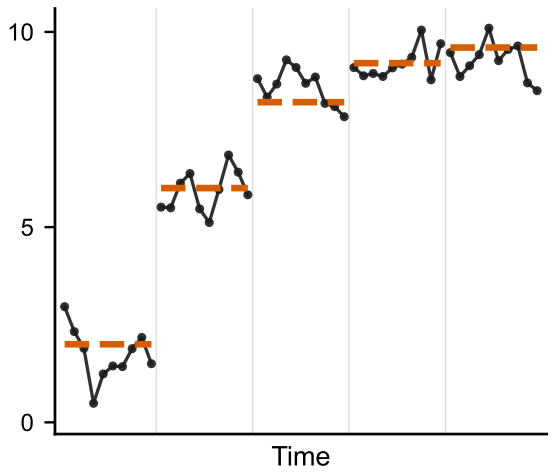
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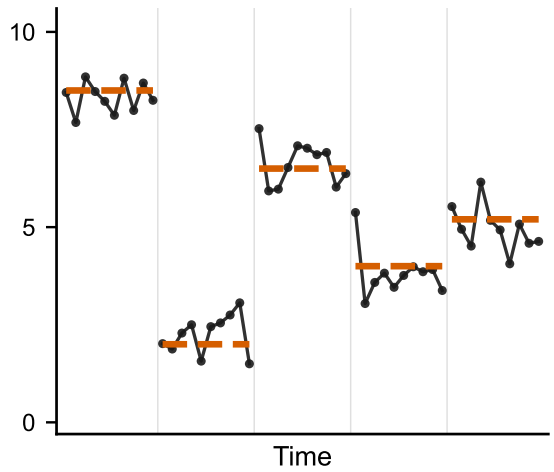
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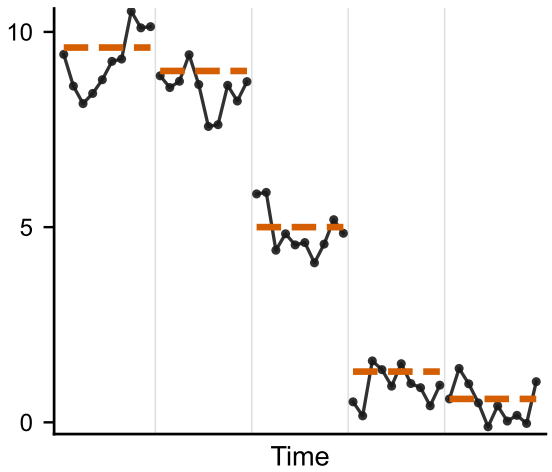
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BurstiVAR, Formally

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$$\boldsymbol{\Phi}_i = \begin{pmatrix} \phi_{11i} & \phi_{12i} \\ \phi_{21i} & \phi_{22i} \end{pmatrix}$$

diagonal = **AR**, off-diagonal = **CR**; the VAR resets at each burst start.

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$B = 1 \Rightarrow$ deviations from a person-specific mean = RI-CLPM (Hamaker et al., 2015)

Bayesian Estimation

- Bayesian estimation via **JAGS** (Plummer, 2003)
- MCMC: 2 chains, adapt $= 5{,}000$, burn-in $= 5{,}000$, iterations $= 20{,}000$
- **Weakly informative priors**
  - AR fixed effects: $N(0,\ 10^2)$ truncated to $(-1.5,\ 1.5)$ to set the scale of the VAR
  - CR fixed effects: $N(0,\ 10^2)$
  - Burst intercepts: $N(0,\ 5^2)$
- Reference benchmarks: $\hat{R} \leq 1.10$ (convergence), ESS ≥ 200 (sampling quality)

Study 1: Minimum Design for Reliable Recovery

Design	VAR Fixed Effects μ_ϕ (worst case)				VAR RE-SDs σ_ϕ
	rBias (%)	Coverage	Power	Type I	rBias range (%)
$T_b=3,\ N=100$	7.9	.94	.39	.05	[+22, +26]
$T_b=3,\ N=500$	4.5	.94	1.00	.05	[+6, +10]
$T_b=5,\ N=100$	2.5	.93	.95	.05	[+10, +15]
$T_b=5,\ N=500$	1.2	.93	1.00	.06	[−4, +3]
$T_b=20,\ N=100$	2.7	.90	1.00	.10	[−3, +1]
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- **Burst-intercept means** μ_α recovered in *every* condition — $|\text{rBias}| \leq 2.5\%$ on the five nonzero means, coverage .90–.98
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BurstiVAR *(unrestricted burst intercepts)*

$\mu_{\alpha_{v1}}, \mu_{\alpha_{v2}}, \ldots, \mu_{\alpha_{vB}}$ estimated freely, no functional form

Per dimension: ***2B*** *params* $=$ *B fixed-effect means* $+$ *B variances (grows with B)*

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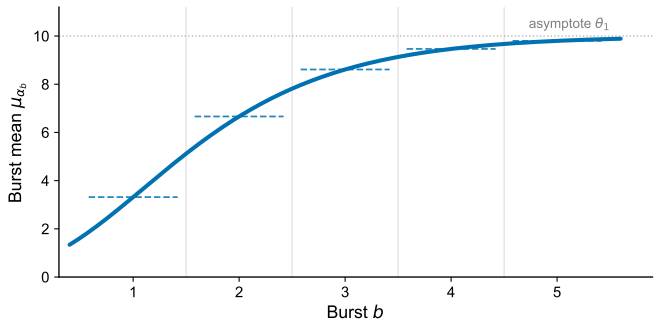
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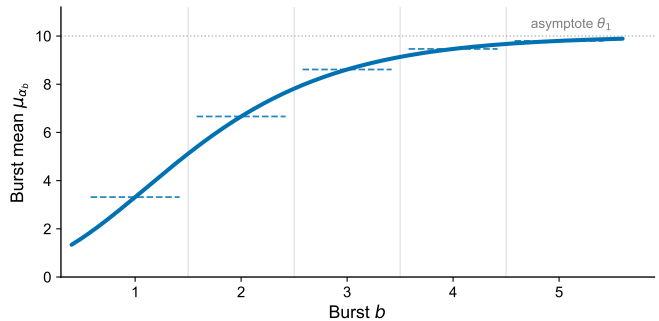
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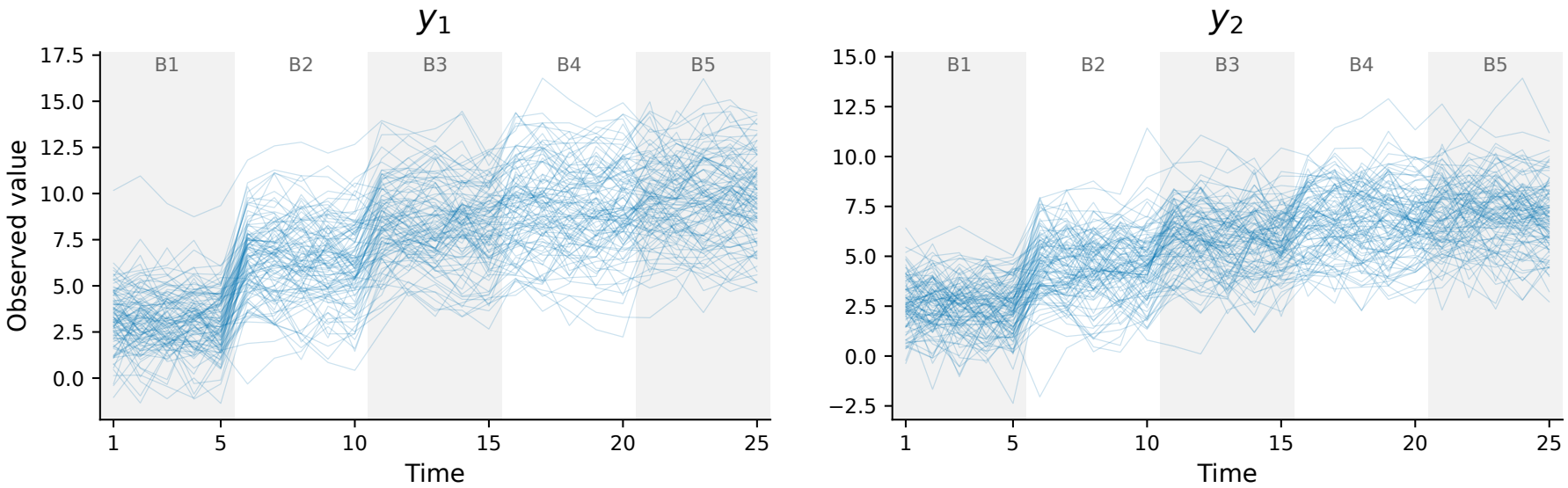
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Q: When the Gompertz restriction holds, what is the efficiency cost of BurstiVAR's extra parameters?

Study 2: Simulated Data (Gompertz Trend)



- Data generated from a *GoBurstiVAR* across $B = 5$ bursts;
- Three conditions $(T_b, N) = (5, 100),\ (5, 500),\ (20, 100)$
- Plots are from one replication of condition $(T_b = 5,\ N = 100)$

Study 2: Trend-Parameter Recovery

- **BurstiVAR** (no Gompertz parameters): its free intercepts recovered the Gompertz-implied burst means and between-person SDs in every condition
  - means $|\text{rBias}| \leq 0.7\%$, SDs $|\text{rBias}| \leq 2.5\%$; coverage .91–.99
- **GoBurstiVAR**: Gompertz fixed effects $|\text{rBias}| \leq 1.8\%$ (coverage .89–.97); asymptote SDs within 1%
- Even the correctly specified GoBurstiVAR could not fully recover the SDs of displacement (σ_{θ_2}) and growth rate (σ_{θ_3}) from $B = 5$ bursts
  - $T_b = 5, N = 100$: σ_{θ_2} $+18\%$ (y_1), $+26\%$ (y_2); σ_{θ_3} (y_2) -15%; coverage down to .75
  - $N \to 500$: biases $\leq 8\%$; $T_b \to 20$ helps only partially

Study 2: Efficiency Cost Findings

- **VAR fixed effects & innovation recovered well** (both specs, all conditions): VAR FEs $|\text{rBias}| \leq 3.1\%$, innovation $\boldsymbol{\Sigma}_{\varepsilon}$ $|\text{rBias}| < 1\%$, coverage $\geq .90$; GoBurstiVAR's RMSE slightly lower at $T_b = 5, N = 100$

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  - ***(largest gap at that corner,*** $T_b = 5, N = 100$***)*** BurstiVAR $[+6, +18]\%$ vs. GoBurstiVAR's $[+1, +5]\%$
  - ***(*** $T_b \to 20$ ***helps a lot)*** BurstiVAR's $|\text{rBias}|$ shrinks to $\leq 2\%$

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  - ***(largest gap at that corner,*** $T_b = 5, N = 100$***)*** BurstiVAR $[+6, +18]\%$ vs. GoBurstiVAR's $[+1, +5]\%$
  - ***(*** $T_b \to 20$ ***helps a lot)*** BurstiVAR's $|\text{rBias}|$ shrinks to $\leq 2\%$
  - ***(*** $N \to 500$ ***helps less)*** BurstiVAR's $|\text{rBias}|$ still up to $+9\%$

Study 2: Efficiency Cost Findings

- **VAR fixed effects & innovation recovered well** (both specs, all conditions): VAR FEs $|\text{rBias}| \leq 3.1\%$, innovation $\boldsymbol{\Sigma}_\varepsilon$ $|\text{rBias}| < 1\%$, coverage $\geq .90$; GoBurstiVAR's RMSE slightly lower at $T_b = 5, N = 100$
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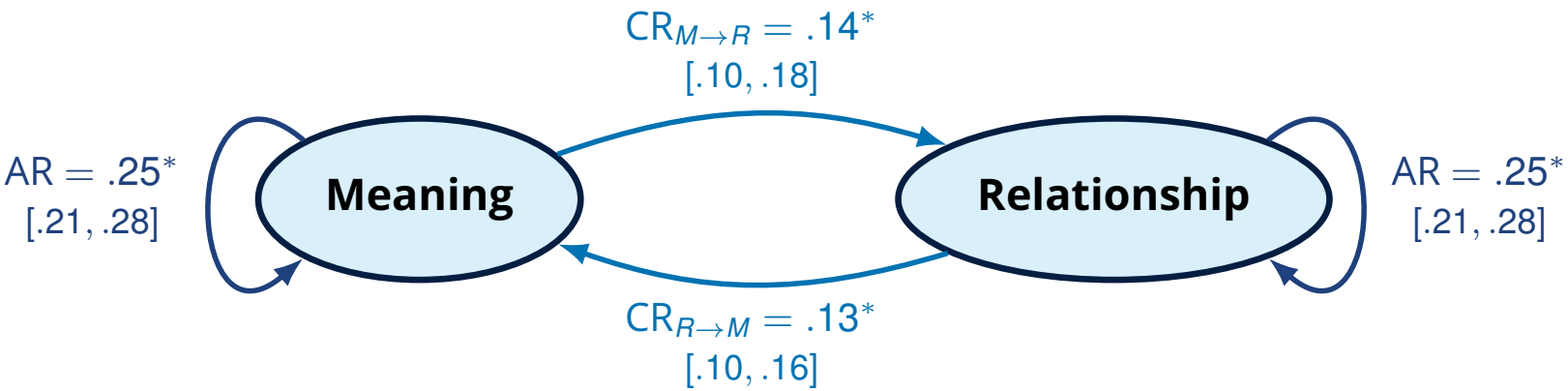
Empirical Example: A Mobile Positive-Psychology Intervention

P1: Pre	**P2: Intv**	**P3: Post-I**	**P4: Post-II**
Days 1–14	Days 15–29	Days 30–42	Days 43–56

4 blocks per day, 1 observation per block $\rightarrow$ $T_b \approx 52$–60 *per phase*

- **Sample:** $N = 108$ participants aged 18–22 (2 of 3 arms); Control (54) vs. PPI+Med (54)
- **Outcomes:** Meaning & Relationship (mPERMA 0–100 slider)
- **Question:** group $\times$ phase shifts *and* within-phase dynamics

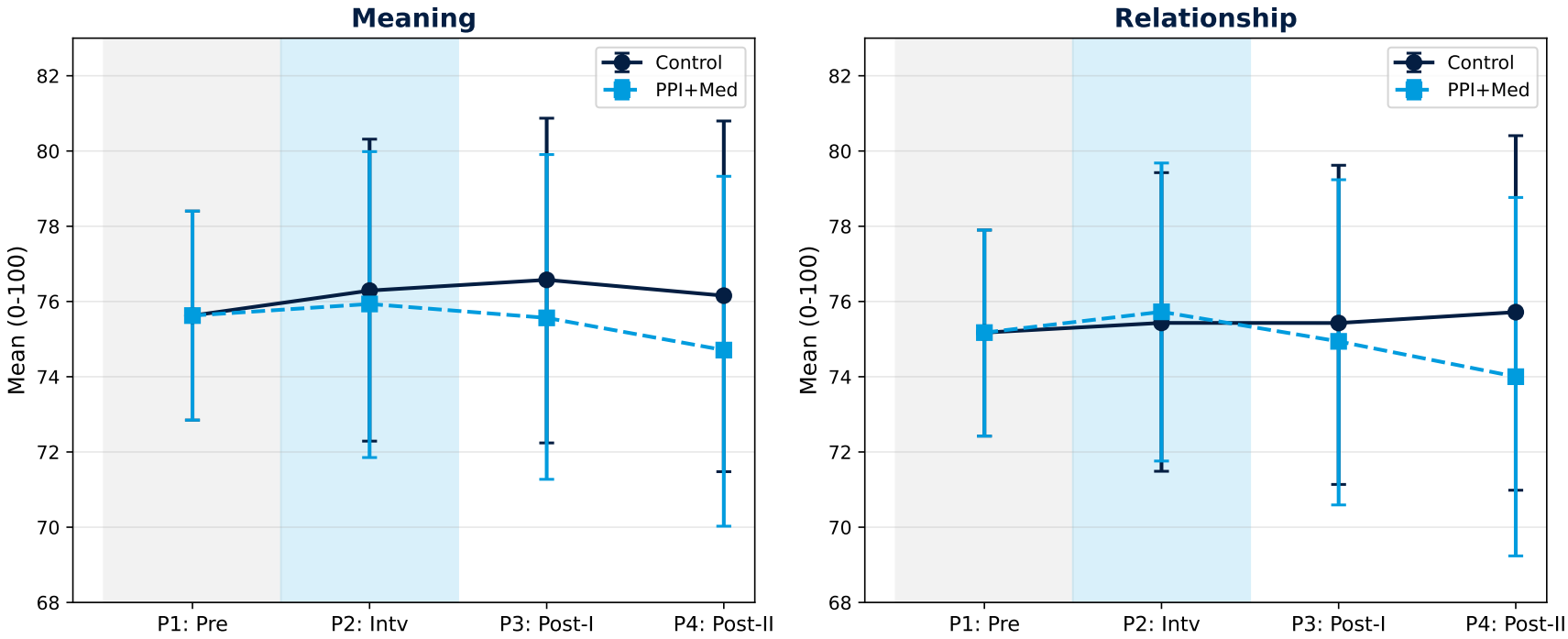
Result I: Bidirectional Positive Coupling



- All four paths: 95% CI excludes zero
- Consistent with Broaden-and-Build (Fredrickson, 2001)

Pooled fit without group covariates. The group-covariate model, which tests group effects on both the phase means and the VAR coefficients, gives AR .23–.26 and CR .11–.17, with no credible group difference.

Result II-a: Phase Means — No Credible Group Differences



- All $\beta_{vb}^{\text{control}}$ (phase) and β_{vb}^{grp} (group) 95% CIs cover zero
- Converges with prior analyses of the same trial (Heshmati et al., 2025; Li et al., 2026)

Result II-b: Within-Burst Sparsity Check

Fit	AR_M	AR_R	$CR_{M\to R}$	$CR_{R\to M}$
Full ($T_b \approx 55$)	**.25**	**.25**	**.14**	**.13**
$T_b = 20$	**.19**	**.24**	**.11**	**.13**
$T_b = 5$	**.35**	**.23**	**.13**	**.19**
$T_b = 3$	**.44**	**.43**	.08 (ns)	**.16**
No burst ($B = 1$)	**.35**	**.32**	**.20**	**.16**

Note. **Bold** = 95% CI excludes zero; (ns) = CI includes zero.

- **At $T_b \geq 5$:** significance and direction preserved on all 4 paths
- **At $T_b = 20$:** estimates within $\pm .06$ of Full
- **No burst ($B = 1$, the RI-CLPM):** all four lagged effects inflated (e.g., AR_M .25 $\rightarrow$ .35)

Summary of Results

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3. **Empirical results:** bidirectional Meaning $\leftrightarrow$ Relationship coupling at the 6-hour lag; no credible group $\times$ phase mean differences; consistent inference of AR and CR when $T_b \geq 5$; ignoring the bursts ($B = 1$) inflates all four lagged effects.

Discussions

- **Two roots:** RI-CLPM's single *time-invariant* intercept (Hamaker et al., 2015) and the local-level model's *free* per-time intercept (Harvey, 1989). BurstiVAR sits between — a **form-free** mean *per burst*, carrying **multivariate** AR/CR dynamics

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- **Restrictions** $\rightarrow$ **next steps:** dynamics invariant across bursts, RE correlations 0, bivariate $\Rightarrow$ evolving dynamics, correlated REs, higher dimension, continuous-time / DSEM

Thank you!


Ivan Jacob A. Pesigan


Zita Oravecz


Stephanie T. Lanza


Sy-Miin Chow

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